

(WASP-39b)

**Project Report Submitted in fulfillment of the
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By

(C. Jane Claudia)
(Sathyabama
Institute of
Science and
Technology)
(145110352)
(509329)

Under the Supervision of
(Dr Ashok Gopalakrishnan)



India Space Academy,

Department of Space Education, India Space Week

1. INTRODUCTION (Exoplanets + WASP-39b)

EXOPLANETS

An exoplanet is a planet that orbits a star outside the solar system. As of February 12, 2026, the NASA Exoplanet Archive confirmed 6,107 total exoplanets and TESS Project Candidates include 7,890 more planets.

Studying exoplanets helps scientists understand planetary formation, atmospheric composition, and habitability. By analyzing the atmospheres of these distant worlds, scientists can detect chemical signatures such as water vapor, methane, and oxygen. These gases not only suggest the possibility of life or habitability but also provide insights into the planets' formation history.

DETECTION METHODS

The two most widely used detection methods are:

- Transit method – This method detects exoplanets by measuring the periodic dipping of a star's brightness, indicating a planet passing in front of it.
- Radial velocity method – This method detects exoplanets by measuring the tiny back-and-forth "wobble" of a star caused by an orbiting planet's gravity.

WASP -39b

WASP-39b is a gas giant exoplanet discovered in February 2011 by the WASP project. It orbits a G8-type star, WASP-39, located about 700 light-years away in Virgo. Similar to our Sun but smaller and cooler, the star hosts WASP-39b, which completes an orbit in approximately 4.055 days. Often classified as a “Hot Saturn” due to its low density, it has a nearly circular orbit and transits its star frequently, allowing detailed study.

WASP-39b became especially important in recent years when carbon dioxide was clearly and unambiguously detected in its atmosphere using the James Webb Space Telescope (JWST). Observations have also identified water vapor and other atmospheric gases, making WASP-39b an important target for studying exoplanet atmospheres.

This project aims to analyze real light curve data of WASP-39b, calculate key physical parameters such as transit depth and planetary radius using transit equations, and interpret the results in terms of the planet's physical characteristics.

2. CORE PHYSICS CALCULATIONS

a) Planet Radius

Formula $R_p = R_* \sqrt{\delta}$

where, R_p = Planet Radius
 R_* = Star Radius
 δ = Transit Depth

$[R_* = 0.939 R_\odot]$
 $[R_\odot = \text{Sun Radius}]$

Transit depth was calculated from the light curve
 $\delta = 0.0268$

$R_* = 0.939 R_\odot$ $[R_\odot = 6.96 \times 10^8 \text{ m}]$
 $R_* = 0.939 \times 6.96 \times 10^8$
 $R_* = 6.53 \times 10^8 \text{ m}$

Substituting values,
 $R_p = 6.53 \times 10^8 \times \sqrt{0.0268}$
 $= 6.53 \times 0.1637 \times 10^8$
 $= 1.068 \times 10^8 \text{ m}$

Planet Radius [WASP-39b] = $1.068 \times 10^8 \text{ m}$

The calculated radius is slightly larger than the literature value. This difference may be due to noise in the light curve or manual estimation of transit depth.

[Actual $R_p = 9.13 \times 10^7 \text{ m}$]

b) Equilibrium Temperature

$$\text{Formula } T_p = T_* \sqrt{\frac{R_*}{2a}}$$

where T_p = Planet Equilibrium Temperature

T_* = Star temperature

R_* = Star Radius

a = Orbital distance

From NASA Exoplanet archive

[Maneini et al 2018]

$$T_* = 5485 \text{ K}$$

DOMS

$$a = 0.4828 \text{ AU}$$

$$[\text{AU} = 1.496 \times 10^{11} \text{ m}]$$

Converting AU to meters,

$$a = 0.4828 \times 1.496 \times 10^{11} \text{ m}$$

$$= 7.22 \times 10^9 \text{ m}$$

$$R_p = 6.53 \times 10^8 \text{ m}$$

[Calculated in planet radius]

Substituting values,

$$T_p = 5485 \sqrt{\frac{6.53 \times 10^8}{2 \times 7.22 \times 10^9}}$$

$$= 5485 \sqrt{\frac{6.53 \times 10^8}{1.44 \times 10^{10}}}$$

$$= 5485 \sqrt{0.0453}$$

$$= 5485 \times 0.2128$$

$$\text{Equilibrium Temperature} = 1167.41 \text{ K} \approx 1170 \text{ K}$$

c) Density

Formula

$$\rho = \frac{3M}{4\pi R^3}$$

M = Mass of planet

R = Planet Radius

[Calculated before]

$$M = 0.28 M_J$$

[M_J = Mass of Jupiter]

$$= 1.898 \times 10^{27} \times 0.28$$

$$= 5.3144 \times 10^{26} \text{ kg}$$

Substituting the values,

$$\rho = \frac{3(5.3144 \times 10^{26})}{4\pi (1.218 \times 10^{24})^3}$$

$$= \frac{1.5943 \times 10^{27}}{1.531 \times 10^{25}}$$

$$= 104 \text{ kg/m}^3$$

$$\text{Density} = 0.104 \text{ g/cm}^3$$

[WASP-39b]

This low density indicates that WASP-39b has a highly extended, gaseous atmosphere and likely a small, low-mass core, suggesting it formed with a high proportion of hydrogen and helium.

The calculated density of WASP-39b is slightly lower than the value reported in published studies. This difference mainly comes from the way the planetary radius was estimated from the transit depth. Since the light curve used in this project was raw observational data and not phase-folded, it contained noise and small fluctuations. The transit depth was determined manually from the minimum flux value, which may introduce minor inaccuracies.

Because density depends on the cube of the radius, even a small overestimation in radius can significantly reduce the calculated density. Therefore, the slightly larger radius obtained in this analysis resulted in a lower density value compared to literature results. Other possible sources of error include instrumental noise, data gaps, and limitations in manual data processing. Despite these uncertainties, the overall result still clearly indicates that WASP-39b is a low-density, inflated gas giant, which is consistent with previously published research.

EARTH VS WASP-39b

PARAMETER	EARTH	WASP-39b
Radius	$6.37 \times 10^6\text{m}$	$1.068 \times 10^8\text{m}$
Mass	$5.97 \times 10^{24}\text{kg}$	$5.314 \times 10^{26}\text{kg}$
Density	5.51 g/cm^3	0.104 g/cm^3
Equilibrium Temperature	288K	1167.41K
Orbital Period	365 days	4.05 days
Type	Rocky Planet(Terrestrial)	Gas Giant
Distance from star (a)	1 AU	0.048 AU

Compared to Earth, WASP-39b is much larger in size and mass but has a significantly lower density, indicating that it is composed mainly of gas rather than solid rock. Its extremely high equilibrium temperature and very short orbital period show that the planet orbits very close to its host star. Because of these conditions, WASP-39b is classified as a hot gas giant and is not considered habitable. The intense heat and lack of a solid surface make it unsuitable for Earth-like life.

DATA WITHOUT CODING

FLUX VS TIME

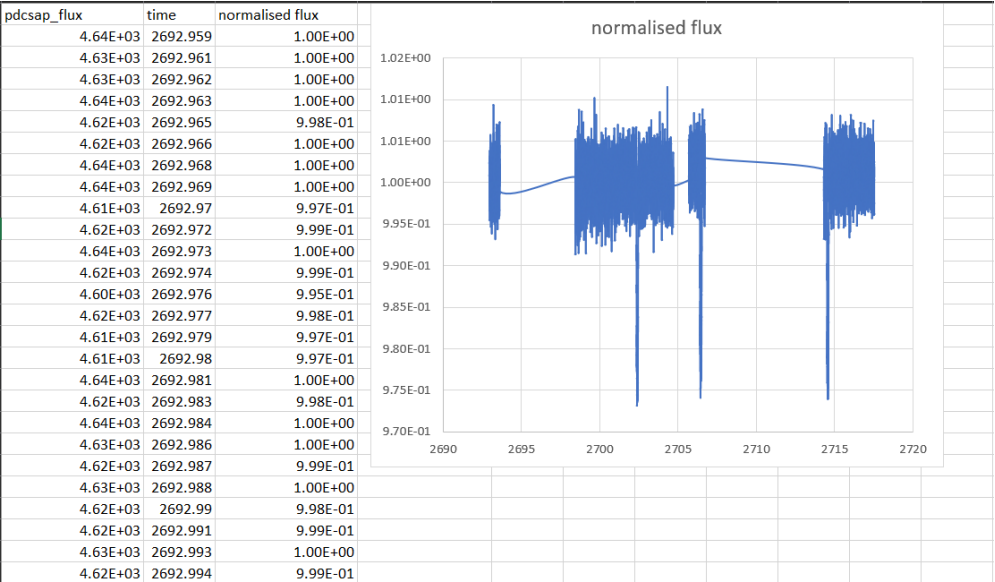


Figure 1: Transit light curve of WASP-39b showing normalized flux vs time.

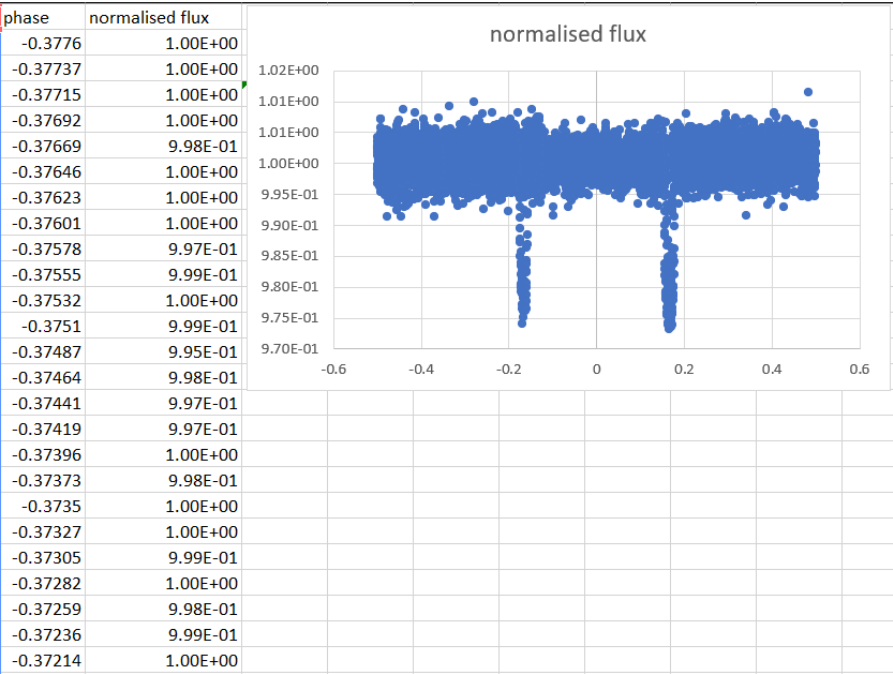


Figure 2: Phase-folded light curve highlighting the transit dip.

Phase Folded Light Curve

To better visualize the transit signal, the data was also plotted using a phase-folded light curve. In this type of graph, the observational data are folded using the planet's orbital period so that multiple transit events overlap. This helps in highlighting the transit signal more clearly.

In the phase-folded plot, most data points cluster around the normalized flux value of 1.0, indicating normal stellar brightness. A clear dip can be seen near the center of the graph, which corresponds to the transit event.

The presence of this dip confirms that the observed decrease in brightness is not random noise but a repeating event caused by the planet passing in front of the star during its orbit.

Light Curve Analysis

The light curve of the host star of WASP-39b was plotted using the normalized flux values obtained from the observational dataset from MAST archive. In the graph of normalized flux versus time, most of the data points remain close to a value of 1.0, which represents the normal brightness of the star when no transit is occurring. However, at certain points in the graph, a noticeable dip in the flux is observed.

This drop in brightness occurs when the planet passes directly in front of the star from the observer's point of view. During this event, a small portion of the starlight is blocked by the planet, causing the observed brightness to decrease temporarily. This phenomenon is known as a **transit** and is one of the most commonly used methods for detecting exoplanets.

The dips observed in the graph clearly indicate the presence of a transit signal. Since the planet orbits the star periodically, these dips appear at regular intervals in the time series data. The depth and shape of these dips provide useful information about the size of the planet relative to the host star.

The minimum normalized flux value obtained from the graph is:

$$\text{Min flux value} = 0.97326$$

The transit depth was estimated manually using:

$$\text{Transit Depth} = 1 - \text{Min flux value}$$

$$\text{Transit Depth} = 1 - 0.97326 = 0.0268$$

This corresponds to a 2.68% decrease in stellar brightness during transit.

The observed dip confirms the transit detection method and allows estimation of planetary radius using the relation:

$$R_p = R_* \sqrt{\delta}$$

Based on the calculated physical parameters, WASP-39b can be clearly classified as a gaseous planet rather than a rocky one. The derived planetary radius is about 1.5 times that of Jupiter, and the calculated density is extremely low (around 0.1 g/cm³) compared to Earth's density of 5.51 g/cm³. Rocky planets generally have high densities because they

are composed of solid materials such as iron and silicates. In contrast, the very low density of WASP-39b indicates that it is composed mainly of light gases such as hydrogen and helium. This confirms that the planet belongs to the category of inflated gas giants.

The equilibrium temperature of WASP-39b was calculated to be approximately 1170 K. Such a high temperature suggests that the planet is located very close to its host star and receives intense stellar radiation. Due to this extreme heat, the planet lies far outside the habitable zone, which is the region around a star where conditions may allow liquid water to exist. Therefore, WASP-39b is not suitable for life as we know it.

Another important feature of WASP-39b is its extended atmosphere. The large planetary radius combined with very low density suggests that the planet has a thick and expanded gaseous envelope. Planets with such low gravity and high temperatures tend to have atmospheres that are easier to observe during transit. This makes WASP-39b a good target for atmospheric studies. In fact, previous observations using space telescopes have detected molecules such as water vapor and carbon dioxide in its atmosphere. These characteristics confirm that the atmosphere of WASP-39b is detectable and scientifically important for understanding exoplanet chemistry and atmospheric processes.

Overall, the derived parameters from this study confirm that WASP-39b is a hot, low-density gas giant located close to its host star, with a large and observable atmosphere.

CONCLUSION

In this project, the physical properties of the exoplanet WASP-39b were studied using real observational data obtained from the MAST archive. The light curve was plotted using flux versus time, and the transit dip was identified manually. From the minimum normalized flux value, the transit depth was calculated to be 0.0268. Using this value and the known stellar radius, the planetary radius was derived. Further calculations were performed to estimate the equilibrium temperature and density of the planet.

Although small differences were observed when compared to published values, the overall results are consistent with established research. This study demonstrates how transit observations combined with basic physics can be used to determine important planetary properties. The project highlights the practical application of astrophysical concepts and provides insight into how astronomers characterize exoplanets using real data.